OmniPay API

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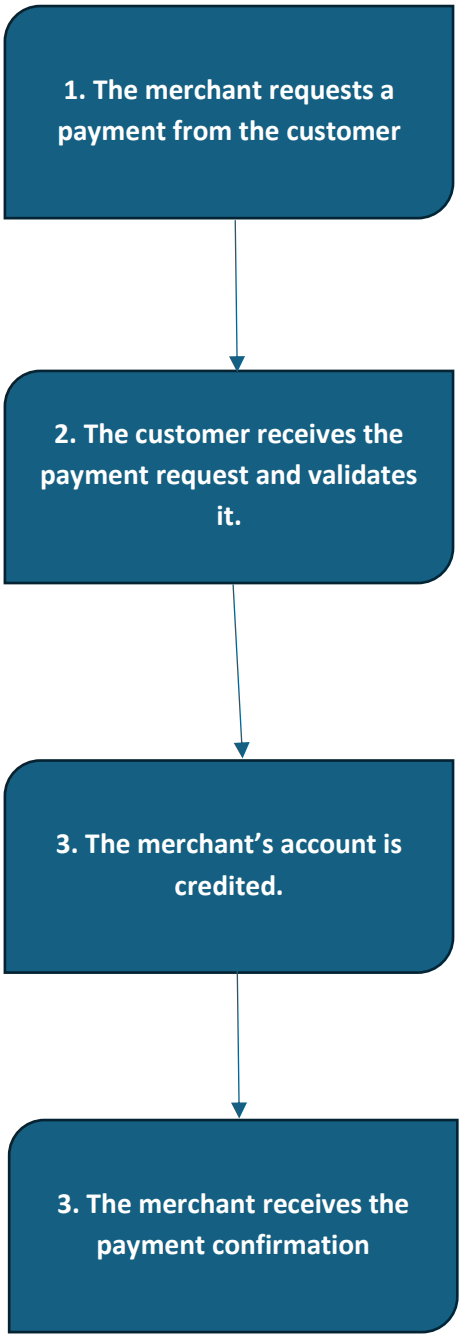
1 Payment process

This process is used to request a payment from a mobile money wallet. It involves the following steps:



2 Payment request

This method is used to request a payment from the customer.

URL : https://omnipay.webtechci.com/interface/api2

Parameter	Description	Mandatory	Example
action	Action to perform	Yes	paymentrequest
apikey	Your API Key	Yes	omnipay_api_key_xxxxxxx
msisdn	Customer's phone number with international prefix without 00 or +	Yes	2250707070707
amount	Amount of the transaction	Yes	1000
reference	A transaction reference generated by the API user (alphanumerical). Must be unique for each transaction.	Yes	REF123456789
first_name	Customer's first name	Yes	John
last_name	Customer's last name	Yes	Doe
otp	One Time Password generated by customer	Yes (only for operators that need an OTP to validate the transaction like ORANGE)	1234
operator	The operator is automatically determined using the customer's phone number (MSISDN). So it is not necessary for Moov, Orange or MTN. But some operators like WAVE can use any type of phone number. In this case it is necessary to specify the operator.	No (Mandatory for wallets like WAVE or MIXX)	Possible values: wave mixx
return_url	The web page were the user will be redirected after payment.	Yes (only for operator WAVE)	https://www.website.com/returnPage.aspx

Example :

```
POST /interface/api2 HTTP/1.1
Host: omnipay.webtechci.com
Content-Type: application/json

{"action":"paymentrequest","apikey":"omnipay_api_key_XXXXXXXXX","msisdn":"2250707000001","amount":"500","reference":"REF123","first_name":"foo","last_name":"bar"}
```

Example for operator WAVE :

```
POST /interface/api2 HTTP/1.1
Host: omnipay.webtechci.com
Content-Type: application/json

{"action":"paymentrequest","apikey":"omnipay_api_key_XXXXXXXXX","msisdn":"2250707000001","amount":"500","reference":"REF123","first_name":"foo","last_name":"bar","operator":"wave","return_url":"https:\/\/www.merchant.com\/confirm.aspx"}
```

The request returns a JSON object with following properties :

Property	Description
success	**"1"** if the request is successful, "**0**" otherwise
code	Error code in case of failure (see next table for error codes)
message	An error message in case of failure
id	The ID of the transaction on Omnipay system in case of success.
reference	User defined reference provided in the request
payment_url	This property is provided only for payment methods that require to follow a link in order to validate the payment such as WAVE. The customer needs to follow this URL in order to validate the payment. After validation the customer will be redirected to the URL provided in parameter "return_url".
first_name	Customer's first name
last_name	Customer's last name
msisdn	Customer's phone number
amount	Amount of the transaction
fees	Fees for the transaction
type	Type of transaction. Value is "PAYMENT" for a payment request.

Examples:

Successful transaction :

```
{"success":1,"type":"PAYMENT","reference":"REF45","id":123456789,"first_name":"foo","last_name":"bar","msisdn":"22507070707","amount":500,"fees":25}
```

Successful transaction for operator WAVE :

```
{"success":1,
,"type":"PAYMENT","reference":"REF45","id":123456789,"payment_url":"htt
```

ps:\/\/pay.wave.com\/c\/cos1xf57psp823h8?a=500&c=XOF&m=omni","first_name":"foo","last_name":"bar","msisdn":"22507070707","first_name":"foo","last_name":"bar","msisdn":"22507070707","amount":500,"fees":25}

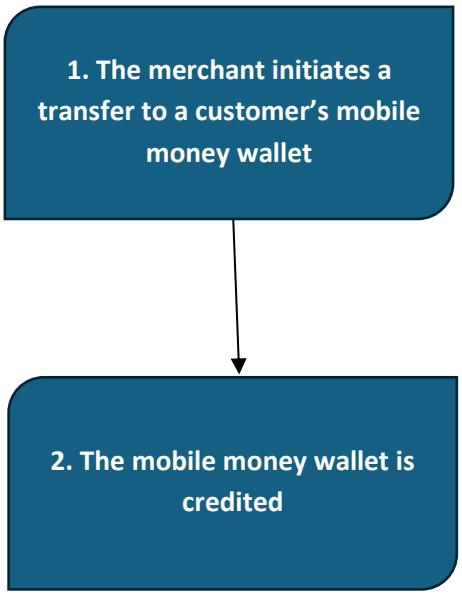
Failed transaction :

{"success":0,"code":1,"message":"Invalid API Key"}

See error codes table for details about error codes.

3 Transfer process

This process is used to send money from the merchant's account to the customer's mobile money wallet.



4 Transfer request

This method is used to send money from the merchant's account to a specified mobile money wallet.

URL : https://omnipay.webtechci.com/interface/api

Parameter	Description	Mandatory	Example
action	Action to perform	Yes	transfer
apikey	Your API Key	Yes	omnipay_api_key_xxxxxxx
msisdn	Customer's phone number with international prefix without 00 or +	Yes	2250707070707
amount	Amount of the transaction	Yes	1000
reference	A transaction reference generated by the API user (alphanumerical). Must be unique for each transaction.	Yes	REF123456789
first_name	Customer's first name	Yes	John

last_name	Customer's last name	Yes	Doe
operator	The operator is automatically determined using the customer's phone number (MSISDN). So it is not necessary for Moov, Orange or MTN. But some operators like WAVE can use any type of phone number. In this case it is necessary to specify the operator.	Yes (only for operator WAVE)	wave

Example :

```
POST /interface/api2 HTTP/1.1
Host: omnipay.webtechci.com
Content-Type: application/json

{"action":"transfer","apikey":"omnipay_api_key_XXXXXXXXX","msisdn":"22507000000001","amount":"300000","reference":"REF1234","first_name":"foo","last_name":"bar"}
```

Example for operator WAVE :

```
POST /interface/api2 HTTP/1.1
Host: omnipay.webtechci.com
Content-Type: application/json

{"action":"transfer","apikey":"omnipay_api_key_XXXXXXXXX","msisdn":"22507000000001","amount":"300000","reference":"REF1234","first_name":"foo","last_name":"bar","operator":"wave"}
```

The request returns a JSON object with following properties :

Property	Value and Description
success	**"1"** if the request is successful, "**0**" otherwise
code	Error code in case of failure (see next table for error codes)
message	An error message in case of failure
id	The ID of the transaction on Omnipay system in case of success.
reference	Reference provided by the user during the request
first_name	Customer's first name
last_name	Customer's last name
msisdn	Customer's phone number
amount	Amount of the transaction
fees	Fees for the transaction
currency	Currency code

type	Type of transaction. Value is "TRANSFER" for a transfer request.

Examples:

Successful transaction :

```
{"success":1,"type":"TRANSFER","reference":"123456","id":123456789,"first_name":"foo","last_name":"bar","msisdn":"22507070707","amount":500,"fees":25,"currency":"XOF"}
```

Failed transaction :

```
{"success":0,"code":1,"message":"Invalid API Key"}
```

5 Get Transaction Status

This method is used to request the status of a transaction.

URL : https://omnipay.webtechci.com /interface/api2

Parameters	Description	Mandatory	Example
action	Action to perform (getstatus)	Yes	getstatus
apikey	Your API Key	Yes	omnipay_api_key_xxxxxxxxxxxxxx
reference	Reference of the transaction as initially defined by the merchant	Yes	REF12345

Example :

```
POST /interface/api2 HTTP/1.1
Host: omnipay.webtechci.com
Content-Type: application/json

{"action":"getstatus","apikey":"omnipay_api_key_XXXXXXXXXXXXXXXXXXXXXXX","reference":"REF6754"}
```

The request returns a JSON object with following properties :

Property	Description
success	**"1"** if the request is successful, **"0"** otherwise
id	Transaction ID
status	1 => Transaction initiated 2 => Pending validation 3 => Success 4 => Transaction failure

reference	Reference of the transaction as defined by the merchant
message	Status description or error message
msisdn	Customer's phone number
amount	Amount of the transaction
fees	Fees for the transaction
type	Type of transaction. Value is "PAYMENT" for a payment request or "TRANSFER" for a transfer request.

Examples

Success


```
{"success":1,"id":1234,"type":"PAYMENT","reference":"12345","status":3,
"message":"Transaction
successful","first_name":"foo","last_name":"bar","msisdn":"22507070707"
,"amount":500,"fees":25}
```


Error


```
{"success":0,"code":19,"message":"Invalid reference"}
```


6 Get transaction status via callback URL

You can specify a callback URL to which the status of transactions will be sent.

Example :


```
POST /callback.php HTTP/1.1
Host: clienthost.com
User-Agent: PHP Script
Content-Type: application/json
Content-Length: 141
Connection: Close

{"action":"callback","id":"170777","type":"PAYMENT","reference":"REF123","first_name":"foo","last_na
me":"bar","msisdn":"2250153958903","amount":"500","fees":"20","currency":"XOF","status":"4","messa
ge":"Transaction failed"}
```


Key	Value and Description
action	callback
id	Transaction ID on Omnipay system
status	1 => Transaction initiated 2 => Pending validation 3 => Transaction successful

	4 => Transaction failure
message	Status description
reference	Reference of the transaction as initially defined by the merchant
amount	Amount of the transaction
fees	Fees of the transaction
currency	Currency of the transaction
first_name	Customer's first name
last_name	Customer's last name
msisdn	Customer's phone number
type	Type of transaction. Value is "PAYMENT" for a payment request or "TRANSFER" for a transfer request.
signature	Signature of the payload to ensure origin and integrity of the payload. (see callback signature)

6.1 Callback signature

To ensure that the callback comes from OMNIPAY and has not been tampered, the payload of the callback is signed using your callback key using HMAC-SHA3-512 algorithm. The hashed payload is sent in the "signature" field. You can compute the payload hash and compare it to the signature field in order to ensure that that callback has not been tampered.

In order to verify the signature you need to concatenate the values of some fields in the callback fields using "|" (pipe) as separator. Then sign the concatenated string with the callback key using HMAC-SHA3-512 algorithm. The callback key used to sign the concatenated string is found in your dashboard "My Account > Callback URL". The signed string must match the value of the "signature" field in the callback.

The fields to concatenate are (in order): id, type, reference, msisdn, amount, fees, status, message

Callback data:

Id=27
type=PAYMENT
reference= REFABC123
amount=500
fees=25
currency=XOF
msisdn=2250505050505
status=4
message=Transaction failed
signature=
f4536e2f0b6fddae1636a3e05a548d076aab40ff29b9fd08482096e6f26ee331ab2e90cd1be8c309e30c5487aa11824a1d8b9ac46fbcff06039c88c7a703b3cb

Concatenated values: 27|PAYMENT|REFABC123|2250505050505|500|25| 4|Transaction failed

Concatenated values signed with callback key using HMAC-SHA3-512 (must match signature value):
f4536e2f0b6fddae1636a3e05a548d076aab40ff29b9fd08482096e6f26ee331ab2e90cd1be8c309e30c5487aa11824a1d8b9ac46fbcff06039c88c7a703b3cb

7 Get balance

This method is used to get the merchant's balance.

URL : https://omnipay.webtechci.com /interface/api2

```
POST /interface/api2 HTTP/1.1
Host: omnipay.webtechci.com
Content-Type: application/json

{"action":"getbalance","apikey":"omnipay_api_key_xxxxxxxxxxxxxxxxxx"}
```

Parameters	Description	Mandatory	Example
action	Action to perform (getbalance)	Yes	getbalance
apikey	Your API Key	Yes	omnipay_api_key_xxxxxx

NB : All parameters must be URL encoded.

The request returns a JSON object with following properties :

Property	Description
success	**"1"** if the request is successful, **"0"** otherwise
code	Error code in case of failure (see error table)
message	An error message in case of failure
balance	An array of objects containing the balance per country. Each object has the following properties:
countryName	Name of the country
countryCode	ISO3 code of the country
amount	Amount of the balance
pending	Amount in pending withdrawals
currency	Currency code

Examples

Success

```
{
  "success": 1,
```

```
  "balance": [
    {
      "countryName": "Ivory Coast",
      "countryCode": "CIV",
      "amount": 26262,
      "currency": "XOF"
    }
  ]
}
```

Error

```
{"success":0,"code":1,"message":"Invalid API Key"}
```

8 Error Codes

Error Code	Description
1	Invalid API Key
2	Service not available
3	Operation not authorized
4	Missing parameter : msisdn
5	Invalid amount
6	Maximum amount exceeded
7	Missing parameter : reference
8	Reference already used
9	Internal communication error
10	Invalid phone number (msisdn)
11	Missing parameter: first_name
12	Missing parameter: last_name
13	Missing parameter: amount
14	Operator not available
15	Minimum amount not reached
16	Invalid MSISDN (phone number)
17	Missing parameter: otp
18	Insufficient funds
19	Invalid reference
20	Missing parameter : return_url
21	Balance not available
22	This MSISDN (phone number) is blacklisted due to fraudulent activity.